

# Research Gap Study

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# Motorcycle lane filtering & maneuvering at signalized approaches

| Paper Title / Source     | APA Reference (7th Ed.)                                                                                                                                                                                                                                                                                                    | Key Focus / Method                                                                                                                                                  | Identified Research Gap in Literature                                                                                                                                                       | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                                                                       | DOI / Reference Link                                                                    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Promraks a et al. (2022) | Promraksa, T., Satiennam, T., Satiennam, W., & Kronprasert, N. (2022). Lane-filtering behavior of motorcycle riders at signalized urban intersections. <i>Journal of Advanced Transportation</i> , 2022, 1–12.                                                                                                             | <b>Onboard measurement device</b> (Lidar, Hall effect, camera) and multilevel linear regression modeling of lateral clearance during filtering.                     | Lack of micro-level empirical data on lateral clearance variations during filtering between stopping vs. moving lateral vehicles at signalized urban intersections in developing countries. | Objective 1               | <b>Uses naturalistic trajectory and drone video</b> data to quantify how lateral clearance, instant filtering speed, neighboring vehicle type, and <b>signal phase govern filtering maneuverability</b> .            | <a href="https://doi.org/10.1155/2022/5662117">https://doi.org/10.1155/2022/5662117</a> |
| Mulvihill et al. (2013)  | Mulvihill, C. M., Salmon, P. M., Filtness, A., Lenné, M., Walker, G. H., Cornelissen, M., & Young, K. L. (2013). Lane filtering and situation awareness in motorcyclists: An on-road proof of concept study. <i>Proceedings of the 2013 Australasian Road Safety Research, Policing &amp; Education Conference</i> , 1–12. | <b>On-road proof-of-concept</b> study examining motorcycle rider situation awareness (SA) during lane filtering at signalized intersections using verbal protocols. | <b>Small sample size</b> (proof-of-concept) and lack of quantitative surrogate safety indicators linking SA degradation directly to conflict risk during queue filtering.                   | Objective 1 & Objective 2 | <b>Integrates</b> naturalistic trajectory observation <b>with surrogate safety measures (TTC, PET, lateral clearance)</b> to quantitatively evaluate the safety impacts of queue filtering at signalized approaches. | <a href="https://eprints.qut.edu.au/63613/">https://eprints.qut.edu.au/63613/</a>       |

## Motorcycle lane filtering & maneuvering at signalized approaches (Contd.)

| Paper Title / Source      | APA Reference (7th Ed.)                                                                                                                                             | Key Focus / Method                                                                                                                                     | Identified Research Gap in Literature                                                                                                                                      | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                                       | DOI / Reference Link                                                                              |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Damani & Vedagiri (2023b) | Damani, J., & Vedagiri, P. (2023). Predicting choice of filtering of motorized two wheelers in urban mixed traffic. Transportation Research Record, 2677(9), 58–68. | Binary logit, Random Forest, and ANFIS modeling of MTW filtering vs. following choices in urban mixed traffic based on lateral gap and relative speed. | <b>Focused primarily on mid-block sections</b> without explicitly accounting for signal phase dynamics, queue accumulation, or specific signalized approach geometry.      | Objective 1               | <b>Extends filtering choice modeling to signalized intersection approaches</b> by incorporating signal phase, queue length, and neighboring vehicle type into the decision criteria. | <a href="https://doi.org/10.1177/03611981231158320">https://doi.org/10.1177/03611981231158320</a> |
| Minh et al. (2012)        | Minh, C. C., Sano, K., & Matsumoto, S. (2012). Maneuvers of motorcycles in queues at signalized intersections. Journal of Advanced Transportation, 46(1), 39–53.    | Maneuverability model framework incorporating dynamic motorcycle lanes, threshold distance, lane selection, and gap acceptance in signalized queues.   | <b>Model restricted to 3 adjacent sub-lanes, relied on manual video extraction</b> , and did not evaluate surrogate safety risk or threshold distances during maneuvering. | Objective 1 & Objective 3 | <b>Utilizes high-resolution trajectory data to capture multi-lane filtering</b> dynamics and calibrates sub-lane simulation models alongside surrogate safety evaluation.            | <a href="https://doi.org/10.1002/atr.144">https://doi.org/10.1002/atr.144</a>                     |

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| Paper Title / Source              | APA Reference (7th Ed.)                                                                                                                                                                                                         | Key Focus / Method                                                                                                                                  | Identified Research Gap in Literature                                                                                                                                         | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                                             | DOI / Reference Link                                                                                                                                                                        |
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| Samalla & Chunchu (2024)          | Samalla, S., & Chunchu, M. (2024). Crash risk associated with powered two-wheeler filtering maneuvers in urban mixed and weak-lane-disciplined traffic streams. <i>Transportmetrica B: Transport Dynamics</i> , 12(1), 2405827. | <b>EVT-based crash risk estimation</b> for filtering vs. non-filtering PTWs using Anticipated Collision Time (ACT) across mid-block urban sections. | Treated PTWs as a homogeneous group without differentiating mopeds/scooters/motorcycles, and <b>ignored signal phase influence and specific lead/adjacent vehicle types</b> . | Objective 1 & Objective 2 | <b>Characterizes filtering behavior</b> across different motorcycle categories, <b>signal phases</b> , and adjacent vehicle types, <b>establishing multi-indicator safety thresholds</b> . | <a href="https://doi.org/10.1080/21680566.2024.2405827">https://doi.org/10.1080/21680566.2024.2405827</a>                                                                                   |
| NSW Centre for Road Safety (2014) | NSW Centre for Road Safety. (2014). Motorcycle lane filtering trial: Summary of trial results. Transport for NSW, Sydney, Australia.                                                                                            | <b>Field evaluation of safety</b> , pedestrian interactions, and rider compliance during a legal <b>motorcycle lane filtering</b> trial in Sydney.  | <b>Macro-level survey</b> and crash analysis lacking microscopic kinematic trajectory data and formal surrogate safety indicators at signalized intersection approaches.      | Objective 1 & Objective 2 | <b>Fills the gap between policy trials and micro-level kinematics</b> by analyzing trajectory-based interactions and surrogate safety measures at signalized approaches.                   | <a href="https://roadsafety.transport.nsw.gov.au/downloads/motorcyclists/lane-filtering-results.pdf">roadsafety.transport.nsw.gov.au/downloads/motorcyclists/lane-filtering-results.pdf</a> |

## Motorcycle lane filtering & maneuvering at signalized approaches (Contd.)

| Paper Title / Source | APA Reference (7th Ed.)                                                                                                                               | Key Focus / Method                                                                                                                     | Identified Research Gap in Literature                                                                                                                                       | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                       | DOI / Reference Link                                                                              |
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| Bella et al. (2024)  | Bella, F., Gulisano, F., & Gagliardi, V. (2024). Safety analysis of motorcyclists' overtaking maneuvers. Accident Analysis & Prevention, 208, 107809. | Hazard-based duration and logistic regression modeling of overtaking duration and critical lateral clearance in suburban environments. | <b>Focused on suburban open-road overtaking</b> using an instrumented car, <b>not addressing queue penetration or lane filtering</b> at signalized intersection approaches. | Objective 1 & Objective 2 | Adapts duration and lateral clearance modeling methodologies specifically to <b>filtering maneuvers and queue penetration at signalized intersection approaches.</b> | <a href="https://doi.org/10.1016/j.aap.2024.107809">https://doi.org/10.1016/j.aap.2024.107809</a> |

# Surrogate safety measures (SSMs) & crash risk assessment for motorcycles

| Paper Title / Source | APA Reference (7th Ed.)                                                                                                                                                                                  | Key Focus / Method                                                                                                                                           | Identified Research Gap in Literature                                                                                                                                           | Target Proposal Obj. | How Proposal Addresses the Gap                                                                                                                                        | DOI / Reference Link                                                                              |
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| Phan et al. (2016)   | Phan, V. L., Evdorides, H., Bradford, J., & Mumford, J. (2016). Motorcycle crash risk models for urban roads. Proceedings of the Institution of Civil Engineers - Transport, 169(6), 397–407.            | Discrete choice and Threshold Safety Distance (TSD) models for rear-end and sideswipe crash potentials in motorcycle-dominated urban traffic.                | <b>Relied on deterministic safety distance assumptions; did not incorporate continuous surrogate indicators (TTC, PET) or signal phase variations</b> at signalized approaches. | Objective 2          | <b>Applies continuous SSMs (TTC, PET, DRAC, lateral-gap indicators)</b> at signalized approaches to establish dynamic conflict thresholds and <b>safety indices</b> . | <a href="https://doi.org/10.1680/jtran.15.00075">https://doi.org/10.1680/jtran.15.00075</a>       |
| Borsos (2021)        | Borsos, A. (2021). Application of bivariate Extreme Value models to describe the joint behavior of temporal and speed related surrogate measures of safety. Accident Analysis & Prevention, 159, 106274. | <b>Bivariate EVT modeling</b> of temporal proximity (TTC, T2) and severity (relative speed, Delta-V) indicators using Generalized Pareto Distribution (GPD). | <b>Focused on general car-following interactions;</b> did not evaluate non-lane-based motorcycle filtering, swerving, or lateral clearance indicators in mixed traffic.         | Objective 2          | <b>Integrates lateral clearance and lateral proximity indicators into bivariate surrogate safety frameworks</b> specifically for motorcycle interactions.             | <a href="https://doi.org/10.1016/j.aap.2021.106274">https://doi.org/10.1016/j.aap.2021.106274</a> |

## Surrogate safety measures (SSMs) & crash risk assessment for motorcycles (Contd.)

| Paper Title / Source     | APA Reference (7th Ed.)                                                                                                                                                                            | Key Focus / Method                                                                                                                    | Identified Research Gap in Literature                                                                                                                         | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                              | DOI / Reference Link                                                                                      |
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| Maji & Ghosh (2025)      | Maji, A., & Ghosh, I. (2025). Advanced traffic conflict analysis for safety evaluation at roundabouts under mixed traffic using extreme value theory. Accident Analysis & Prevention, 219, 108108. | <b>Bivariate Peak over Threshold (POT) EVT modeling at roundabouts</b> using TTC and MaxDeltaV, coupled with SSAM conflict detection. | Analysis restricted to roundabout geometries; does not model motorcycle-specific lane filtering or queue-discharge conflicts at signalized approaches.        | Objective 2               | <b>Applies bivariate EVT modeling to signalized intersection approaches</b> , focusing on motorcycle filtering and queue penetration interactions.          | <a href="https://doi.org/10.1016/j.aap.2025.108108">https://doi.org/10.1016/j.aap.2025.108108</a>         |
| Damani & Vedagiri (2024) | Damani, J., & Vedagiri, P. (2024). An investigation of following behavior and associated safety of MTWs in heterogeneous traffic. Transportation Letters, 16(5), 439–451.                          | Generalized Linear Modeling (GLM) of distance/time gaps and safety analysis using TTC and PICUD for strict and oblique following.     | <b>Focused primarily on mid-block following regimes; did not examine conflict risk during queue filtering</b> under red/green signal phases at intersections. | Objective 1 & Objective 2 | <b>Evaluates oblique following and filtering conflicts specifically at signalized intersection approaches</b> under varying signal phases and queue states. | <a href="https://doi.org/10.1080/19427867.2023.2203995">https://doi.org/10.1080/19427867.2023.2203995</a> |



## Surrogate safety measures (SSMs) & crash risk assessment for motorcycles (Contd.)

| Paper Title / Source          | APA Reference (7th Ed.)                                                                                                                                                                     | Key Focus / Method                                                                                                                                     | Identified Research Gap in Literature                                                                                                                      | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                      | DOI / Reference Link                                                                                    |
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| Avathkattil & Vedagiri (2026) | Avathkattil, S., & Vedagiri, P. (2026). Empirical evaluation of vehicle safety envelope across vehicle type and speed in disordered traffic condition. IATSS Research, 50, 633–652.         | <b>Mathematical modeling of two-dimensional Vehicle Safety Envelope (VSE)</b> based on speed and vehicle type in disordered traffic.                   | VSE parameters assumed constant multi-vehicle influence without isolating pairwise motorcycle filtering interactions or signal phase transition effects.   | Objective 1 & Objective 2 | <b>Refines the 2D Vehicle Safety Envelope concept to evaluate dynamic lateral and longitudinal clearances during motorcycle filtering at signalized approaches.</b> | <a href="https://doi.org/10.1016/j.iatssr.2025.12.006">https://doi.org/10.1016/j.iatssr.2025.12.006</a> |
| Zheng & Sayed (2019)          | Zheng, L., & Sayed, T. (2019). Comparison of traffic conflict indicators for crash estimation using <b>peak over threshold approach</b> . Transportation Research Record, 2673(5), 493–502. | <b>Performance evaluation of univariate POT models across multiple conflict indicators</b> (TTC, PET, Extended TTC) for intersection crash estimation. | <b>Evaluated standard car traffic conflicts; did not account for non-lane-based motorcycle maneuvers or lateral clearance surrogate safety indicators.</b> | Objective 2               | <b>Incorporates motorcycle-specific lateral surrogate safety measures alongside temporal proximity indicators in POT extreme value crash risk estimation.</b>       | <a href="https://doi.org/10.1177/0361198119841556">https://doi.org/10.1177/0361198119841556</a>         |

## Surrogate safety measures (SSMs) & crash risk assessment for motorcycles (Contd.)

| Paper Title / Source | APA Reference (7th Ed.)                                                                                                                                                                                                                                         | Key Focus / Method                                                                                                                                  | Identified Research Gap in Literature                                                                                                                                 | Target Proposal Obj. | How Proposal Addresses the Gap                                                                                                                                 | DOI / Reference Link                                                                              |
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| Chen et al. (2024)   | Chen, Y., Bian, Y., Yuan, Q., King, M., He, J., Cao, X., ... & Zheng, Y. (2026). Exploring novel surrogate safety indicators measuring conflict riskiness and severity: a case study in Sacramento, United States. Accident Analysis & Prevention, 228, 108400. | <b>Partially elastic collision theory for conflict severity and multi-indicator selection across straight-line and angle conflicts.</b>             | <b>Restricted to highway straight-line and angle car conflicts;</b> explicitly noted inability to model vulnerable road users or non-lane-based motorcycle filtering. | Objective 2          | <b>Develops multi-indicator conflict severity metrics tailored to motorcycle-vehicle interactions and filtering maneuvers</b> at signalized approaches.        | <a href="https://doi.org/10.1016/j.aap.2026.108400">https://doi.org/10.1016/j.aap.2026.108400</a> |
| Lu et al. (2020)     | Lu, C., He, X., van Lint, H., Tu, H., Happee, R., & Wang, M. (2020). Performance evaluation of surrogate measures of safety with naturalistic driving data. Accident Analysis & Prevention, 148, 105794.                                                        | Comparative analysis of 6 SMOs (TTC, DRAC, PICUD, S-PDRF, PFS, CFS) on timeliness, accuracy, precision, and robustness using naturalistic car data. | <b>Strictly focused on car rear-end conflicts; excluded steering/swerving evasive actions</b> and motorcycle non-lane-based maneuvers.                                | Objective 2          | <b>Evaluates SMOs performance specifically for motorcycle maneuvers</b> involving combined braking and lateral swerving at signalized intersection approaches. | <a href="https://doi.org/10.1016/j.aap.2021.106403">https://doi.org/10.1016/j.aap.2021.106403</a> |

## Surrogate safety measures (SSMs) & crash risk assessment for motorcycles (Contd.)

| Paper Title / Source   | APA Reference (7th Ed.)                                                                                                                                                                                             | Key Focus / Method                                                                                                                                | Identified Research Gap in Literature                                                                                                                       | Target Proposal Obj. | How Proposal Addresses the Gap                                                                      | DOI / Reference Link                                                                                                                            |
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| Fazekas et al. (2017)  | Fazekas, A., Hennecke, F., Kalló, E., & Oeser, M. (2017). A novel surrogate safety indicator based on constant initial acceleration and reaction time assumption. Journal of Advanced Transportation, 2017, 1–9.    | Formulation of comprehensive time-based conflict indicators <b>integrating TET, TIT, and TTC</b> under constant initial acceleration assumptions. | <b>Formulated exclusively for strict single-lane car-following scenarios;</b> fails under non-lane-based mixed traffic with motorcycle filtering/swerving.  | Objective 2          | Accommodate lateral displacement and dynamic gap acceptance during <b>motorcycle filtering</b> .    | <a href="https://doi.org/10.1155/2017/8376572">https://doi.org/10.1155/2017/8376572</a>                                                         |
| St-Aubin et al. (2015) | St-Aubin, P., Saunier, N., & Miranda-Moreno, L. F. (2015). Comparison of various time-to-collision prediction and aggregation methods for surrogate safety analysis. Transportation Research Record, 2520, 155–162. | Comparison of motion prediction methods (constant velocity, constant acceleration) for TTC calculation in automated safety analysis.              | <b>Tested primarily on roundabouts and highway interchanges for four-wheeled vehicles;</b> lacked validation on motorcycle filtering and queue penetration. | Objective 2          | <b>Motorcycle trajectory dynamics at signalized intersection approaches</b> to refine TTC accuracy. | <a href="http://n.saunier.free.fr/saunier/stock/st-aubin14methodology.pdf">http://n.saunier.free.fr/saunier/stock/st-aubin14methodology.pdf</a> |

## Surrogate safety measures (SSMs) & crash risk assessment for motorcycles (Contd.)

| Paper Title / Source | APA Reference (7th Ed.)                                                                                                                                                     | Key Focus / Method                                                                                                                       | Identified Research Gap in Literature                                                                                                     | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                      | DOI / Reference Link                                                          |
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| Zhao & Lee (2017)    | Zhao, P., & Lee, C. (2017). Analysis and validation of surrogate safety measures by types of lead and following vehicles. Transportation Research Record, 2659(1), 137–147. | Trajectory-based evaluation of SSM distributions across different lead-following vehicle configurations (Car-Car, Car-Truck, Truck-Car). | <b>Did not include motorized two-wheelers (MTWs) as lead or following vehicles</b> , ignoring the dominant vehicle type in mixed traffic. | Objective 1 & Objective 2 | Explicitly <b>analyzes vehicle-pair combinations involving motorcycles (MTW-Car, MTW-HV, MTW-MTW)</b> to establish <b>vehicle-type-specific safety thresholds</b> . | <a href="https://doi.org/10.3141/2659-15">https://doi.org/10.3141/2659-15</a> |

# Microscopic traffic simulation & sub-lane behavioral modeling

| Paper Title / Source                               | APA Reference (7th Ed.)                                                                                                                                                                                                   | Key Focus / Method                                                                                                                              | Identified Research Gap in Literature                                                                                                                                                                                   | Target Proposal Obj. | How Proposal Addresses the Gap                                                                                                                                                    | DOI / Reference Link                                                                                        |
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| Wichitpho W. Wichitphongsa, W., ngsa et al. (2025) | Kronprasert, N., Zaw, M. S., Pisetsit, P., & Satiennam, T. (2025). Modeling riding and stopping behaviors at motorcycle box intersections: A case study in Chiang Mai City, Thailand. <i>Infrastructures</i> , 10(4), 97. | <b>External Driver Model (EDM)</b> embedded in PTV <b>VISSIM</b> <b>simulating motorcycle filtering</b> and box intersection stopping behavior. | <b>Focused</b> primarily on peak-hour traffic at <b>motorcycle box intersections</b> ; did not calibrate sub-lane behavior for standard signalized approaches without boxes or across varying signal phase transitions. | Objective 3          | <b>Calibrates</b> microscopic simulation models of <b>sub-lane behavior</b> and <b>lateral positioning</b> across diverse signalized approach layouts and signal phase sequences. | <a href="https://doi.org/10.3390/infrastructure10040097">https://doi.org/10.3390/infrastructure10040097</a> |
| Lee et al. (2009)                                  | Lee, T. C., Polak, J. W., & Bell, M. G. H. (2009). New approach to modeling mixed traffic containing motorcycles in urban areas. <i>Transportation Research Record</i> , 2140(1), 195–205.                                | <b>Agent-based simulation (Bikesim)</b> combining headway models, path choice, and Gipps car-following to model motorcycle movements.           | <b>Calibrated using mid-block arterial data</b> from London; <b>lacked validation on high-density motorcycle queue discharge and filtering at signalized approaches</b> in developing cities.                           | Objective 3          | Calibrates sub-lane simulation models against empirical trajectory data from motorcycle-dense signalized intersection approaches.                                                 | <a href="https://doi.org/10.3141/2140-22">https://doi.org/10.3141/2140-22</a>                               |

# Microscopic traffic simulation & sub-lane behavioral modeling (Contd.)

| Paper Title / Source | APA Reference (7th Ed.)                                                                                                                                                                                              | Key Focus / Method                                                                                                                       | Identified Research Gap in Literature                                                                                                                        | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                            | DOI / Reference Link                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Minh et al. (2006)   | Minh, C. C., Sano, K., & Matsumoto, S. (2006). Models of motorcycle deceleration behavior at signalized intersections. <i>Infrastructure Planning Review</i> , 23, 927–936.                                          | Free-deceleration and following-deceleration models of motorcycles during red lights using dynamic motorcycle lanes.                     | <b>Focused solely on longitudinal deceleration during red intervals; omitted acceleration, lateral maneuvering, and queue discharge during green phases.</b> | Objective 1 & Objective 3 | Develops a unified simulation framework covering both red-phase queue buildup/deceleration and green-phase queue filtering/acceleration.  | <a href="https://doi.org/10.2208/journalip.23.927">https://doi.org/10.2208/journalip.23.927</a> |
| Minh et al. (2007)   | Minh, C. C., Sano, K., & Matsumoto, S. (2007). Acceleration and deceleration models of motorcycle at signalized intersections. <i>Journal of the Eastern Asia Society for Transportation Studies</i> , 7, 2728–2741. | Parameter estimation of acceleration and deceleration regimes based on longitudinal threshold distances and reaction time distributions. | Data collected from limited intersection approaches; <b>did not simulate lateral sub-lane positioning or filtering choices</b> across adjacent lanes.        | Objective 3               | <b>Incorporates 2D sub-lane lateral positioning and filtering choices</b> into the calibrated acceleration/deceleration simulation model. | <a href="https://doi.org/10.11175/easts.7.2396">https://doi.org/10.11175/easts.7.2396</a>       |

# Microscopic traffic simulation & sub-lane behavioral modeling (Contd.)

| Paper Title / Source | APA Reference (7th Ed.)                                                                                                                                                                     | Key Focus / Method                                                                                                                                              | Identified Research Gap in Literature                                                                                                               | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                     | DOI / Reference Link                                                                          |
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| Hsu & Chen (2014)    | Hsu, T., & Chen, Y. (2014). Mobility force model for microscopic simulation of motorcycle traffic behavior. IEEE Conference Proceedings on Intelligent Transportation Systems, 1–6.         | <b>Conceptual 2D force field model (attraction and repulsion forces) for motorcycle</b> car-following and space perception.                                     | Theoretical force-field formulation lacking rigorous empirical calibration against high-resolution trajectory data at signalized approaches.        | Objective 3               | <b>Uses empirical trajectory data</b> from signalized intersection approaches to <b>calibrate sub-lane behavioral</b> parameters and interactive force boundaries. | <a href="https://doi.org/10.2991/msam-15.2015.11">https://doi.org/10.2991/msam-15.2015.11</a> |
| Trinh et al. (2017)  | Trinh, L. T., Sano, K., & Colleagues. (2018). <b>Two-player game theory</b> based analysis of <b>motorcycle</b> drivers' behavior at intersections. Asian Transport Studies, 5(2), 272–291. | Game-theoretic modeling of ' <b>cut tail</b> ' and ' <b>giving way</b> ' interactions for left-turning <b>motorcycles</b> using <b>agent-based simulation</b> . | Focused on turning conflict points; <b>did not address longitudinal queue filtering or lateral clearance maintenance on signalized approaches</b> . | Objective 1 & Objective 3 | Sub-lane filtering models to simulate queue penetration and lateral positioning on signalized approaches.                                                          | <a href="https://doi.org/10.11175/eastsats.5.272">https://doi.org/10.11175/eastsats.5.272</a> |



# Microscopic traffic simulation & sub-lane behavioral modeling (Contd.)

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| Huynh (2016)         | Huynh, H. D. N. (2016). Capacity analysis of signalised intersections in motorcycle dependent cities (Doctoral dissertation). Technische Universität Darmstadt, Darmstadt, Germany.                         | Saturation flow rate and capacity modeling for motorcycle-dependent cities (MDCs) in Vietnam considering non-lane-based flow. | <b>Macro-level capacity and saturation flow formulation; lacked micro-level sub-lane trajectory simulation</b> and individual filtering behavior modeling. | Objective 3               | Bridges the gap between micro-level sub-lane filtering trajectories and macro-level intersection approach capacity through calibrated microscopic simulation. | <a href="https://doi.org/10.25534/tuprints-00011474">https://doi.org/10.25534/tuprints-00011474</a>     |
| Guido et al. (2011)  | Guido, G., Astarita, V., Giofré, V., & Vitale, A. (2011). Safety performance measures: A comparison between microsimulation and observational data. Procedia - Social and Behavioral Sciences, 20, 217–225. | SSAM-based safety evaluation using simulated vehicle trajectories vs. field observed traffic conflicts.                       | <b>Tested standard car traffic in SSAM; failed to capture non-lane-based motorcycle filtering or swerving conflicts</b> in simulation output.              | Objective 2 & Objective 3 | <b>Calibrates sub-lane simulation</b> output to produce realistic non-lane-based motorcycle trajectories for surrogate safety evaluation.                     | <a href="https://doi.org/10.1016/j.sbspro.2011.08.027">https://doi.org/10.1016/j.sbspro.2011.08.027</a> |



## Microscopic traffic simulation & sub-lane behavioral modeling (Contd.)

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| Habtemichael & de Picado-Santos (2014) | Habtemichael, F. G., & de Picado-Santos, L. (2013). The impact of high-risk drivers and benefits of limiting their driving degree of freedom. Accident Analysis & Prevention, 60, 305-315. | <b>VISSIM and SSAM simulation modeling</b> of high-risk driving behaviors (speeding, tailgating, abrupt lane changes). | <b>Modeled highway four-wheeled vehicle drivers</b> ; did not consider vulnerable road users, motorcycles, or signalized approach filtering. | Objective 2 & Objective 3 | <b>Incorporates motorcycle-specific risky maneuvers</b> (aggressive filtering, tight lateral gaps) into calibrated intersection approach simulations. | <a href="https://doi.org/10.1016/j.aap.2013.05.013">https://doi.org/10.1016/j.aap.2013.05.013</a> |

# Video trajectory analysis, kinematic data & high-risk event detection

| Paper Title / Source | APA Reference (7th Ed.)                                                                                                                                                                                                       | Key Focus / Method                                                                                                                    | Identified Research Gap in Literature                                                                                                                  | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                                | DOI / Reference Link                                                                                      |
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| Jain et al. (2024)   | Jain, A., & Colleagues. (2024). Investigating spatiotemporal traffic dynamics toward conflict risk levels using trajectory data in heterogeneous traffic. Transportation Research Part C: Emerging Technologies, 165, 104720. | <b>UAV trajectory extraction,</b> machine learning feature selection (RF, XGBoost, LightGBM), and Ordered Probit risk level modeling. | <b>Focused on divided 6-lane highways;</b> did not examine signalized intersection approach dynamics, signal phase changes, or queue filtering.        | Objective 1 & Objective 2 | Applies UAV trajectory extraction and spatiotemporal feature selection specifically to <b>signalized intersection approaches under varying signal phases.</b>                 | <a href="https://doi.org/10.1080/15389588.2025.2541269">https://doi.org/10.1080/15389588.2025.2541269</a> |
| Wu et al. (2020)     | Wu, Y., Abdel-Aty, M., Cai, Q., Lee, J., & Park, J. (2020). Automated safety diagnosis based on unmanned aerial vehicle video and deep learning algorithm. Transportation Research Record, 2674(10), 350–360.                 | Deep learning (Mask R-CNN + CSRT tracking) for UAV trajectory extraction and PET conflict heatmap generation.                         | <b>Tested primarily on car traffic at intersections;</b> did not validate detection/tracking accuracy for small, densely packed filtering motorcycles. | Objective 1 & Objective 3 | Utilizes high-precision computer vision algorithms to <b>extract micro-level motorcycle trajectory data</b> for <b>filtering characterization</b> and simulation calibration. | <a href="https://doi.org/10.1177/0361198120925808">https://doi.org/10.1177/0361198120925808</a>           |

# Video trajectory analysis, kinematic data & high-risk event detection (Contd.)

| Paper Title / Source  | APA Reference (7th Ed.)                                                                                                                                                                                                  | Key Focus / Method                                                                                                           | Identified Research Gap in Literature                                                                                                                                              | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                                | DOI / Reference Link                                                                                |
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| Kluger & Smith (2016) | Kluger, R., & Smith, B. L. (2016). Identification of safety-critical events using kinematic vehicle data. Accident Analysis & Prevention, 96, 162–168.                                                                   | Time-series pattern matching on single-vehicle longitudinal acceleration from SHRP2 NDS to detect crashes and near-crashes.  | <b>Restricted to light four-wheeled vehicles using single-axis acceleration</b> ; failed to capture motorcycle lateral swerving or multi-vehicle spatial interactions.             | Objective 2               | <b>Combines longitudinal kinematic data with lateral clearance</b> indicators and spatial trajectory data to identify unsafe motorcycle maneuvers.            | <a href="https://doi.org/10.1016/j.aap.2016.08.006">https://doi.org/10.1016/j.aap.2016.08.006</a>   |
| Cafiso et al. (2014)  | Cafiso, S., Di Graziano, A., Giudice, O., & Pappalardo, G. (2014). Using GPS data to detect critical events in motorcycle rider behaviour. International Journal of Mobile Network Design and Innovation, 5(4), 195–204. | GPS speed/position data, cubic smoothing splines, and Mahalanobis distance outlier detection for motorcycle critical events. | <b>Low-cost GPS data suffered from spatial inaccuracy and lacked surrounding vehicle context</b> , making it impossible to evaluate filtering lateral gaps or conflict indicators. | Objective 1 & Objective 2 | Overcomes single-vehicle GPS limitations by <b>integrating video/UAV trajectory data</b> to capture surrounding vehicle context and exact lateral clearances. | <a href="https://doi.org/10.1504/IJMNDI.2014.067182">https://doi.org/10.1504/IJMNDI.2014.067182</a> |

# Video trajectory analysis, kinematic data & high-risk event detection (Contd.)

| Paper Title / Source   | APA Reference (7th Ed.)                                                                                                                                                                                                          | Key Focus / Method                                                                                                               | Identified Research Gap in Literature                                                                                                                             | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                               | DOI / Reference Link                                                                                                      |
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| Koshizen et al. (2020) | Koshizen, T., Sato, F., Oishi, R., & Yamakawa, K. (2021, July). Predicting motorcycle riding behavior using vehicle density variation. In 2021 IEEE Intelligent Vehicles Symposium Workshops (IV Workshops) (pp. 114-121). IEEE. | <b>Smartphone-based algorithm predicting high-risk motorcycle pore-seeking</b> (lateral gap) acceleration in dense traffic flow. | Simplified pore-seeking heuristic for smartphone apps; lacked microscopic trajectory validation or formal surrogate safety threshold estimation at intersections. | Objective 1 & Objective 2 | Provides rigorous <b>trajectory-based analysis of lateral gap acceptance</b> and establishes validated surrogate safety thresholds at signalized approaches. | <a href="https://doi.org/10.1109/IVWorkshops54471.2021.9669216">https://doi.org/10.1109/IVWorkshops54471.2021.9669216</a> |
| Gu et al. (2019)       | Gu, X., Abdel-Aty, M., Xiang, Q., Cai, Q., & Yuan, J. (2019). Utilizing UAV video data for in-depth analysis of drivers' crash risk at interchange merging areas. Accident Analysis & Prevention, 123, 159–169.                  | UAV trajectory extraction and real-time crash risk modeling at freeway interchange merging sections.                             | <b>Focused on highway merging areas for passenger cars</b> ; did not investigate urban signalized intersection approaches or non-lane-based motorcycle filtering. | Objective 1 & Objective 2 | Applies UAV video trajectory extraction methodology to urban signalized approaches to <b>analyze motorcycle filtering</b> and conflict risk.                 | <a href="https://doi.org/10.1016/j.aap.2018.11.010">https://doi.org/10.1016/j.aap.2018.11.010</a>                         |

# Video trajectory analysis, kinematic data & high-risk event detection (Contd.)

| Paper Title / Source       | APA Reference (7th Ed.)                                                                                                                                                                                                                                        | Key Focus / Method                                                                                                                       | Identified Research Gap in Literature                                                                                                                  | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                            | DOI / Reference Link                                                                              |
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| Ezzati Amini et al. (2022) | Ezzati Amini, R., Yang, K., & Antoniou, C. (2022). Development of a conflict risk evaluation model to assess pedestrian safety in interaction with vehicles. Accident Analysis & Prevention, 175, 106773.                                                      | Logistic regression conflict risk evaluation models for pedestrian-vehicle interactions based on minimum distance and conflicting speed. | <b>Focused on pedestrian-vehicle interactions;</b> ignored vehicle-motorcycle and motorcycle-filtering interactions at signalized approaches.          | Objective 2               | Adapts multi-parameter conflict risk evaluation modeling to <b>motorcycle-vehicle interactions</b> and filtering maneuvers at signalized approaches.      | <a href="https://doi.org/10.1016/j.aap.2022.106773">https://doi.org/10.1016/j.aap.2022.106773</a> |
| Attal et al. (2015)        | Attal, F., Boubezoul, A., Samé, A., & Oukhellou, L. (2015, April). <b>Powered-Two-Wheeler safety critical events</b> recognition using a mixture model with quadratic logistic proportions. In Proc. 23rd Eur. Symp. Artif. Neural Netw.(ESANN) (pp. 421-426). | Machine learning classification (fall, near fall, naturalistic riding) using instrumented motorcycle sensors.                            | <b>Focused on single-vehicle loss-of-control (falls); omitted multi-vehicle traffic conflict analysis</b> and lane filtering at signalized approaches. | Objective 1 & Objective 2 | Evaluates safety-critical events arising from multi-vehicle interactions, <b>lateral proximity</b> , and queue <b>filtering</b> at signalized approaches. | <a href="https://hal.science/hal-01215028/">https://hal.science/hal-01215028/</a>                 |

# Video trajectory analysis, kinematic data & high-risk event detection (Contd.)

| Paper Title / Source     | APA Reference (7th Ed.)                                                                                                                                                                                                        | Key Focus / Method                                                                                           | Identified Research Gap in Literature                                                                                                            | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                                                | DOI / Reference Link                                                                              |
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| Maji & Ghosh (2022)      | Maji, A., & Ghosh, I. (2025). Advanced traffic conflict analysis for <b>safety evaluation at roundabouts</b> under mixed traffic using extreme value theory. Accident Analysis & Prevention, 219, 108108.                      | Drone-based video trajectory analysis for proactive safety evaluation and SSM extraction.                    | <b>Evaluated general intersection/roundabouts at car conflicts; did not address non-lane-based motorcycle filtering</b> or sub-lane positioning. | Objective 1 & Objective 2 | Uses drone/UAV trajectory data specifically to extract non-lane-based motorcycle filtering trajectories and evaluate surrogate conflict risk. | <a href="https://doi.org/10.1016/j.aap.2025.108108">https://doi.org/10.1016/j.aap.2025.108108</a> |
| Abdelraouf et al. (2022) | Abdelraouf, A., Abdel-Aty, M., Wang, Z., & Zheng, O. (2022). Trajectory prediction for vehicle conflict identification at intersections using sequence-to-sequence recurrent neural networks. arXiv preprint arXiv:2210.08009. | Deep learning (RNN/LSTM) for <b>real-time trajectory conflict prediction</b> in multi-lane weaving sections. | <b>Developed for car weaving on freeways;</b> lacked adaptation to chaotic motorcycle filtering trajectories at signalized approaches.           | Objective 2 & Objective 3 | <b>Calibration of motorcycle sub-lane simulation</b> at signalized approaches.                                                                | <a href="https://doi.org/10.48550/arXiv.2210.08009">https://doi.org/10.48550/arXiv.2210.08009</a> |

## Video trajectory analysis, kinematic data & high-risk event detection (Contd.)

| Paper Title / Source | APA Reference (7th Ed.)                                                                                                                                                            | Key Focus / Method                                                                                       | Identified Research Gap in Literature                                                                      | Target Proposal Obj.      | How Proposal Addresses the Gap                                                                                            | DOI / Reference Link                                                          |
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| Zhao & Lee (2017)    | Zhao, P., & Lee, C. (2017). Analysis and <b>validation of surrogate safety measures</b> by types of lead and following vehicles. Transportation Research Record, 2659(1), 137–147. | Trajectory-based evaluation of SSM distributions across different lead-following vehicle configurations. | <b>Omitted motorcycles from vehicle-pair</b> taxonomy, ignoring VRU interaction dynamics in mixed traffic. | Objective 1 & Objective 2 | Establishes trajectory-derived SSM distributions for vehicle pairs <b>involving motorcycles</b> at signalized approaches. | <a href="https://doi.org/10.3141/2659-15">https://doi.org/10.3141/2659-15</a> |